LOCALIZATION YIELDS AN EQUIVALENCE MOD_A / T -> MOD_{S^{-1}A} FOR THE SERRE SUBCATEGORY OF S-TORSION MODULES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file examples.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: Fullness is witnessed by the same essential-surjectivity isomorphism; the author proves faithfulness and essential surjectivity and states the descended functor without a separate fullness claim - content verbatim.

Editorial note. The label proposition-localization-and-serre-quotients names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own proposition environment in the chapter source.

Allowed standard prerequisites (editorial).

- Serre subcategories and quotient categories of abelian categories; the universal property of the quotient functor (Homology, serre-subcategory-is-kernel) and the faithful/quotient-by-kernel criterion (Homology, quotient-by-kernel-exact-functor)
- Exactness of localization (Algebra, proposition-localization-exact) and S^{-1}M congruent to M tensor S^{-1}A (Algebra, tensor-localization)

Source reference. Original lemma 00DK: sources/stacks-chow/tags.txt, source lines 500–500. Prerequisite pointer invoked at 6805; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original proposition 00CS: sources/stacks-chow/tags.txt, source lines 472–472. Prerequisite pointer invoked at 6795; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 06XK: sources/stacks-chow/tags.txt, source lines 8854–8854. Prerequisite pointer invoked at 6806; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 02MS: sources/stacks-chow/tags.txt, source lines 3455–3455. Prerequisite pointer invoked at 6798; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Date: stacks-project:examples.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Proposition 0.1. *Let A be a ring. Let S be a multiplicative subset of A. Let Mod_A denote the category of A-modules and $\mathcal{T}$ its Serre subcategory of modules for which any element is annihilated by some element of S. Then there is a canonical equivalence $Mod_A/\mathcal{T} \to Mod_{S^{-1}A}$.*

Proof. The functor $\mathrm{Mod}_A \to \mathrm{Mod}_{S^{-1}A}$ given by $M \mapsto M \otimes_A S^{-1}A$ is exact (by Algebra, Proposition 00CS) and maps modules in $\mathcal{T}$ to zero. Thus, by the universal property given in Homology, Lemma 02MS, the functor descends to a functor $\mathrm{Mod}_A/\mathcal{T} \to \mathrm{Mod}_{S^{-1}A}$.

Conversely, any A-module M with $M \otimes_A S^{-1}A = 0$ is an object of $\mathcal{T}$, since $M \otimes_A S^{-1}A \cong S^{-1}M$ (Algebra, Lemma 00DK). Thus Homology, Lemma 06XK shows that the functor $\mathrm{Mod}_A/\mathcal{T} \to \mathrm{Mod}_{S^{-1}A}$ is faithful.

Furthermore, this embedding is essentially surjective: a preimage to an $S^{-1}A$-module N is N_A, that is N regarded as an A-module, since the canonical map $N_A \otimes_A S^{-1}A \to N$ which maps $x \otimes a/s$ to $(a/s) \cdot x$ is an isomorphism of $S^{-1}A$-modules. □